20 Practical AI Business Analysis Case Studies

This practical guide contains 20 real-world Business Analysis case studies from multiple industries. Each case study demonstrates how Business Analysts can practically use AI for requirement gathering, process analysis, API analysis, data mapping, UAT, documentation, stakeholder communication, automation, and solution validation.

Case Study 1 – Digital Customer Onboarding Modernization

Domain: Retail Banking

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the retail banking domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a digital customer onboarding modernization solution.*
• *Analyze the business process for digital customer onboarding modernization and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for digital customer onboarding modernization.*
• *Create UAT test cases for the digital customer onboarding modernization workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 2 – Healthcare Claims Automation

Domain: Healthcare

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the healthcare domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures

• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a healthcare claims automation solution.*
• *Analyze the business process for healthcare claims automation and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for healthcare claims automation.*
• *Create UAT test cases for the healthcare claims automation workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 3 – Insurance Claim Fraud Detection

Domain: Insurance

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the insurance domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:

• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a insurance claim fraud detection solution.*
• *Analyze the business process for insurance claim fraud detection and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for insurance claim fraud detection.*
• *Create UAT test cases for the insurance claim fraud detection workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 4 – Payment Gateway Retry Handling

Domain: Payments

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the payments domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements

• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a payment gateway retry handling solution.*
• *Analyze the business process for payment gateway retry handling and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for payment gateway retry handling.*
• *Create UAT test cases for the payment gateway retry handling workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 5 – AI Product Recommendation Engine

Domain: E-commerce

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the e-commerce domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a ai product recommendation engine solution.*
• *Analyze the business process for ai product recommendation engine and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for ai product recommendation engine.*
• *Create UAT test cases for the ai product recommendation engine workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications

• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 6 – Phone Bill Payment Mobile App

Domain: Telecommunications

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the telecommunications domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a phone bill payment mobile app solution.*
• *Analyze the business process for phone bill payment mobile app and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for phone bill payment mobile app.*
• *Create UAT test cases for the phone bill payment mobile app workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 7 – Reference Data Modernization

Domain: Capital Markets

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the capital markets domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a reference data modernization solution.*
• *Analyze the business process for reference data modernization and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for reference data modernization.*
• *Create UAT test cases for the reference data modernization workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 8 – Portfolio Reporting Data Services

Domain: Wealth Management

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the wealth management domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a portfolio reporting data services solution.*
• *Analyze the business process for portfolio reporting data services and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for portfolio reporting data services.*
• *Create UAT test cases for the portfolio reporting data services workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 9 – Fleet Tracking and Yard Optimization

Domain: Logistics

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the logistics domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a fleet tracking and yard optimization solution.*
• *Analyze the business process for fleet tracking and yard optimization and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for fleet tracking and yard optimization.*
• *Create UAT test cases for the fleet tracking and yard optimization workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases

• Stakeholder Analysis
• Risk and Dependency Log

Case Study 10 – Inventory Forecasting and Supply Chain Visibility

Domain: Retail

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the retail domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a inventory forecasting and supply chain visibility solution.*
• *Analyze the business process for inventory forecasting and supply chain visibility and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for inventory forecasting and supply chain visibility.*
• *Create UAT test cases for the inventory forecasting and supply chain visibility workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 11 – Student Admission Portal Enhancement

Domain: Education

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the education domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a student admission portal enhancement solution.*
• *Analyze the business process for student admission portal enhancement and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for student admission portal enhancement.*
• *Create UAT test cases for the student admission portal enhancement workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 12 – Flight Booking Cancellation Automation

Domain: Travel

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the travel domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a flight booking cancellation automation solution.*
• *Analyze the business process for flight booking cancellation automation and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for flight booking cancellation automation.*
• *Create UAT test cases for the flight booking cancellation automation workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 13 – Predictive Maintenance Dashboard

Domain: Manufacturing

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the manufacturing domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a predictive maintenance dashboard solution.*
• *Analyze the business process for predictive maintenance dashboard and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for predictive maintenance dashboard.*
• *Create UAT test cases for the predictive maintenance dashboard workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications

• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 14 – Employee Referral and Rewards Platform

Domain: HRMS

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the hrms domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a employee referral and rewards platform solution.*
• *Analyze the business process for employee referral and rewards platform and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for employee referral and rewards platform.*
• *Create UAT test cases for the employee referral and rewards platform workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 15 – Citizen Complaint Management System

Domain: Government

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the government domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a citizen complaint management system solution.*
• *Analyze the business process for citizen complaint management system and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for citizen complaint management system.*
• *Create UAT test cases for the citizen complaint management system workflow.*

Practical Hands-On Exercises:

• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 16 – Mortgage Loan Origination System

Domain: Real Estate

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the real estate domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a mortgage loan origination system solution.*
• *Analyze the business process for mortgage loan origination system and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for mortgage loan origination system.*
• *Create UAT test cases for the mortgage loan origination system workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 17 – Smart Meter Consumption Analytics

Domain: Energy

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the energy domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a smart meter consumption analytics solution.*
• *Analyze the business process for smart meter consumption analytics and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for smart meter consumption analytics.*
• *Create UAT test cases for the smart meter consumption analytics workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases

• Stakeholder Analysis
• Risk and Dependency Log

Case Study 18 – Clinical Trial Data Validation

Domain: Pharmaceutical

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the pharmaceutical domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a clinical trial data validation solution.*
• *Analyze the business process for clinical trial data validation and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for clinical trial data validation.*
• *Create UAT test cases for the clinical trial data validation workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 19 – Order Tracking and Delivery Optimization

Domain: Food Delivery

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the food delivery domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a order tracking and delivery optimization solution.*
• *Analyze the business process for order tracking and delivery optimization and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for order tracking and delivery optimization.*
• *Create UAT test cases for the order tracking and delivery optimization workflow.*

Practical Hands-On Exercises:

• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

Case Study 20 – Security Incident Response Automation

Domain: Cybersecurity

Project Overview:
The organization is facing operational inefficiencies, delays, manual processing, data inconsistencies, and customer dissatisfaction in the cybersecurity domain. Leadership wants to modernize the process using AI-enabled digital solutions.

Key Stakeholders:
• Business Users
• Product Owner
• Operations Team
• Developers
• QA Team
• Compliance Team
• Customers/End Users

Business Analyst Responsibilities:
• Conduct stakeholder interviews and workshops
• Create AS-IS and TO-BE process flows
• Document business and functional requirements
• Perform API and system integration analysis
• Analyze source-to-target data mappings
• Prepare user stories and acceptance criteria
• Support UAT and defect triage
• Perform impact analysis for downstream systems

How AI Can Be Practically Used:
• Generate requirement questionnaires
• Summarize meeting transcripts
• Create BRD and FRD drafts
• Generate user stories and acceptance criteria
• Generate SQL validation queries
• Create UAT scenarios and edge cases
• Analyze API payloads and response structures
• Identify missing requirements and process gaps

Sample AI Prompts:
• *Act as a Senior Business Analyst and generate user stories for a security incident response automation solution.*
• *Analyze the business process for security incident response automation and identify automation opportunities.*
• *Generate API integration requirements and edge case scenarios for security incident response automation.*
• *Create UAT test cases for the security incident response automation workflow.*

Practical Hands-On Exercises:
• Prepare a BRD for the solution
• Create process flow diagrams
• Generate API mapping documents
• Prepare a stakeholder analysis matrix
• Create UAT test cases
• Identify business risks and dependencies

Expected Deliverables:
• BRD / FRD
• User Stories and Acceptance Criteria
• Process Flows
• Data Mapping Documents
• API Requirement Specifications
• UAT Test Cases
• Stakeholder Analysis
• Risk and Dependency Log

How to Use These Case Studies:

1. Pick one domain every week.
2. Practice requirement gathering using AI.
3. Generate deliverables such as BRDs, user stories, API requirements, and UAT cases.
4. Use tools like ChatGPT, Claude, Gemini, Jira, Confluence, Postman, SQL, and diagramming tools.
5. Build your portfolio using these practical exercises.

The future Business Analyst will not only gather requirements but will also leverage AI to accelerate analysis, documentation, validation, testing, and delivery.